

Liver & Hepatobiliary Imaging

The Big Picture

How US, CT and MRI show the liver, gallbladder, biliary tree, spleen and pancreas — and the **signature imaging findings** that separate cysts, abscesses, haemangiomas, metastases and primary tumours. Classify lesions as **diffuse vs focal**, then **cystic / complex / solid**.

Normal Appearances

- **US:** uniform texture, bright portal-triad echoes, echo-free hepatic/portal veins.
- **CT:** liver density **65 ± 5 HU** (higher than muscle/spleen); normal intrahepatic ducts not seen.
- **MRI:** liver **higher signal than spleen on T1**, lower on T2; problem-solver after US/CT (vessels best on coronal).

Multiple Liver Lesions — DDX

Metastases, abscesses, haemangiomas, regenerating nodules, hydatid cysts, simple cysts, primary tumour (HCC).

- **Cystic:** simple cyst, hydatid, cystic mets.
- **Complex:** benign/malignant, inflammatory/neoplastic; haematoma can look complex; hard to call unless clearly a simple cyst.

Liver Metastases (commoner than primary)

- **Multiple, variable size, peripheral.**
- **US:** usually **hypoechoic**, but can be hyperechoic, complex, cystic, **target**, or isoechoic.
- **CT:** rounded, lower density than liver; **ring/lesion enhancement** on CECT.
- **MRI:** excellent — **low T1, high T2**, ring enhancement.

Primary Malignant (HCC, cholangiocarcinoma)

Similar imaging to metastases but **often large, usually solitary** (may be multifocal).

Benign Lesions

- **Simple cyst** — US: thin/invisible wall, echo-free, **posterior enhancement**; CT: **0 HU water density**, well-defined (<1 cm indistinguishable from tumour); MRI low T1/high T2. Congenital; multiple → **ADPKD**.
- **Hydatid cyst** — **wall calcification, visible wall, daughter cysts**.
- **Haemangioma** (common, incidental) — US: strongly **echogenic**; CT: low density with **characteristic centripetal (peripheral→central) enhancement** → fills in; MRI: **very high T2 "light bulb"** (like cyst).
- **Abscess** — US: thick/irregular wall, internal debris, chronic calcifies; **aspiration under US**; CT: thick wall, fluid centre (higher density than cyst); resembles necrotic tumour.

Trauma

- **CT best** (detects other organ injury, intraperitoneal blood).
- Commonest: parenchymal **contusion/laceration** + subcapsular haematoma (low density; high-density clot may be seen).

Cirrhosis & Portal Hypertension

- **Commonest cause of portal HTN = cirrhosis** (also veno-occlusive disease, portal vein thrombosis).
- Cirrhosis: abnormal texture (US early, CT later), **right lobe atrophy**, irregular outline.
- Portal HTN: **splenomegaly, collaterals**; US/Doppler, CT, MRV. **TIPS** (interventional) has largely replaced surgery.

Fatty Liver (NAFLD/MAFLD)

- Common; diffuse (can be focal); **2nd leading indication for liver transplant in USA**.
- **US: bright liver; CT: low density; MRI: high T1** (fat signal) — MRI helpful in doubt.

Walk-Away Points

- Liver density on CT = **65 ± 5 HU**, brighter than spleen; MRI: liver > spleen on T1.
- **Metastases**: multiple, hypoechoic, ring-enhancing, low T1/high T2.
- **Haemangioma**: centripetal fill-in + **T2 "light bulb"**; **hydatid**: wall + daughter cysts.
- **CT best for liver trauma; cirrhosis = top cause of portal HTN** (splenomegaly + collaterals, treat with TIPS).

Liver & Hepatobiliary Imaging — Revision Table	
Modality	Normal liver
US	Uniform, bright portal triads
CT	65 ± 5 HU , > muscle/spleen
MRI	> spleen on T1 , < on T2
Lesion class	DDx
Cystic	Simple cyst, hydatid, cystic mets
Complex	Abscess, necrotic tumour, haematoma
Solid	Mets, HCC, haemangioma
Metastases	Imaging
Pattern	Multiple, peripheral, variable
US	Usually hypoechoic / target
CT	Ring enhancement, lower density
MRI	Low T1, high T2
Benign lesion	Signature
Simple cyst	0 HU , echo-free, posterior enhancement
Hydatid	Wall calcification + daughter cysts
Haemangioma	Centripetal fill-in + T2 light-bulb
Abscess	Thick irregular wall + debris, aspirate
Primary malignant	Large, usually solitary (HCC, cholangio)
Other	Finding
Trauma	CT best ; contusion/laceration + subcapsular haematoma
Cirrhosis	RT lobe atrophy, irregular outline
Portal HTN	Cirrhosis commonest; splenomegaly, collaterals, TIPS
Fatty liver	Bright US, low CT, high T1 MRI

Bold = high-yield.